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gayathri@Gayathri: ~/system x + v
gayathri@Gayathri:~$ sudo apt update
[sudo: authenticate] Password:
Hit:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:4 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
107 packages can be upgraded. Run 'apt list --upgradable' to see them.
gayathri@Gayathri:~$ sudo apt install gedit gcc strace
gedit is already the newest version (48.1-9build1).
gcc is already the newest version (4:15.2.0-5ubuntu1).
gcc set to manually installed.
strace is already the newest version (6.19+ds-0ubuntu5).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 107
gayathri@Gayathri:~$ mkdir -p ~/system_call_program
gayathri@Gayathri:~$ cd ~/system_call_program
gayathri@Gayathri:~/system_call_program$ gedit main.c

** (gedit:633): WARNING **: 07:00:51.271: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:633): WARNING **: 07:00:51.271: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
gayathri@Gayathri:~/system_call_program$ gedit copyfile.c

** (gedit:1276): WARNING **: 07:01:21.886: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:1276): WARNING **: 07:01:21.886: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
gayathri@Gayathri:~/system_call_program$ gedit file.c

** (gedit:2078): WARNING **: 07:02:32.211: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:2078): WARNING **: 07:02:32.212: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
gayathri@Gayathri:~/system_call_program$ cd ~/system_call_program

gayathri@Gayathri:~/system_call_program$ cd ~/system_call_program
gayathri@Gayathri:~/system_call_program$ mkdir -p ~/system_call_program
cd ~/system_call_program
gayathri@Gayathri:~/system_call_program$ cat > copyfile.c
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <stdlib.h>

int main()
{
    int source, destination;
    char buffer[1024];
    ssize_t bytesRead, bytesWritten;

    source = open("sample.txt", O_RDONLY);

    if (source < 0)
    {
        perror("Error opening source file");
        return 1;
    }

    destination = open("copy.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (destination < 0)
    {
        perror("Error opening destination file");
        close(source);
        return 1;
    }

    while ((bytesRead = read(source, buffer, sizeof(buffer))) > 0)
    {
        bytesWritten = write(destination, buffer, bytesRead);
    }
}
```



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gayathri@Gayathri: ~/system
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\3\0>\0\1\0\0\0 \250\2\0\0\0\0"... , 832) = 832
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libpcre2-8.so.0", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0\0\0"... , 832) = 832
close(3) = 0
openat(AT_FDCWD, "/proc/filesystems", O_RDONLY|O_CLOEXEC) = 3
read(3, "nodev\tsysfs\nnodev\ttmpfs\nnodev\tbd"... , 1024) = 442
close(3) = 0
openat(AT_FDCWD, "/proc/mounts", O_RDONLY|O_CLOEXEC) = 3
read(3, "none /usr/lib/modules/6.18.33.2-"... , 1024) = 1024
read(3, "ime 0 \0\ndevpts /dev/pts devpts r"... , 1024) = 1024
read(3, "x5,access=client,msize=65536,tra"... , 1024) = 1024
read(3, "nix tmpfs ro,relatime 0 \0\nnone /"... , 1024) = 597
read(3, "", 1024) = 0
close(3) = 0
openat(AT_FDCWD, "/proc/self/maps", O_RDONLY|O_CLOEXEC) = 3
read(3, "637236461000-637236553000 r--p 0"... , 1024) = 1024
read(3, "0 08:30 13647 "... , 1024) = 1024
read(3, "00 08:30 13650 "... , 1024) = 1024
read(3, " /usr/lib/x86_64-lin"... , 1024) = 1024
read(3, "0 13644 /us"... , 1024) = 566
close(3) = 0
openat(AT_FDCWD, "/usr/share/coreutils/locales/uucore/en-US.ftl", O_RDONLY|O_CLOEXEC) = 3
read(3, "# Common strings shared across a"... , 4080) = 4080
read(3, "", 32) = 0
close(3) = 0
openat(AT_FDCWD, "/usr/share/coreutils/locales/cat/en-US.ftl", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "sample.txt", O_RDONLY|O_CLOEXEC) = 3
This is a sample file.
This file is being copied using system calls.close(5) = 0
close(4) = 0
close(3) = 0
+++ exited with 0 +++
gayathri@Gayathri:~/system_call_program$

```